

FAQ on the Data Management module

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Below are reported clarifications regarding the numerous questions relating to the data management module (DM below). For further questions, please contact Prof. Andrea Maurino

Faq 1. Exam procedutes

The DM exam consists of two parts: a written one and a project discussed during the oral session. Both are assigned a maximum grade of 30/30 cum laude.

The final grade of DM is given by the formula: $\text{written_grade} \cdot 0.5 + \text{oral_grade} \cdot 0.5$

Faq 2. Conduct of the exam

The two parts of the exam can be taken in two different sessions of the same academic year.

Faq 3. Written exam

The written exam consists of open-ended questions. it is possible to answer in English or in Italian. The final vote is expressed in the thirtieths. The written exam is considered passed if the candidate obtains an overall mark equal to or greater than 15/30.

Faq 4. Project: objectives, composition, and characteristics

The DM project aims to evaluate the candidates' ability to answer "research questions" on a domain chosen by the candidates using the technologies shown during the course. The project is normally carried out by two or three candidates; exceptions are allowed to be agreed upon with the teacher

The project must cover all phases of a typical data science pipeline (data acquisition, storage, profiling, integration, analysis, and quality improvement phases).

The topic of the DM project is freely chosen by the candidates, but it is necessary that each group agrees with the teacher on the topic and sources used.

The project is presented by the entire group that developed it during the oral sessions.

Faq 5. Data Acquisition

It is mandatory to identify at least two data sources, different from each other, and for at least one of the two sources either scraping techniques or the use of APIs must be used.

Faq 6. Storage

the integrated data must be modeled and stored in one or more DBMS (relational, document-based, graph...)

It is necessary to make at least two queries on the data

Faq 7. Data integration

For the implementation of the DM project, it is necessary to integrate/enrich at least two different datasets. The project must show how the integration and/or enrichment activity shown during the course was automated, measuring the errors related to the integration/enrichment project. It is up to

the group to use existing tools or develop custom solutions for integration. It is suggested to pay particular attention to cases of integration/spatial enrichment of datasets.

Faq 9. Data profiling/data quality

it is necessary to evaluate either the quality of the data obtained after the acquisition phase or the quality of the data before and after the integration/enrichment activity.

It is up to the group to choose the most appropriate dimensions and metrics for measuring data quality

Faq 8. Project Documentation and Submission

Candidates who intend to submit the project must share all the documentation with the teacher by the date of the written exam in which they intend to present the project using a Google Drive folder. The folder must contain a ppt presentation of the project, a report in pdf format, and all the code developed, as well as the data and an operational guide that contains all the information useful for the reproducibility of the project.

The report, in English, must clearly indicate:

- The components of the project and their role
- The research questions that the project wants to try to solve
- The datasets used and the method of acquisition
- The process performed or the phases of (depending on the specific project) acquisition, storage, exploratory analysis, cleaning, integration, enrichment,
- The software architecture used
- Conclusions, possible improvements and/or future developments

All technical choices must be adequately commented and it is necessary to be consistent with the choices made.

The report is subject to evaluation

Faq 9. Oral discussion

During the oral exams, candidates will present the project by preparing a 15-minute presentation. It is up to the group how to organize the presentation, but all members must present a part of the project.

The project is considered passed if it obtains an overall score equal to or greater than 15/30. in the event of failure, a new project must be submitted